CSP-J 2025 (Junior)

2025-10-25, 8:30–12:00 (3.5 hours)

This file is tailored to Luogu (these problems are not on QOJ): every problem reads from standard input and writes to standard output (the original paper used the files `<code>.in` / `<code>.out` named after each problem).
In every subtask table, a cell that reads "as above" repeats the entry of the row above it (the original tables merge such cells).

Problem 1: Forming a Number (拼数) Luogu P14357

Time limit: 1.0 s. Memory limit: 512 MB.

Little R is learning string processing. Little X gives Little R a string s, which consists only of lowercase English letters and digits, and which contains at least one digit among $1 \sim 9$. Little X wants Little R to take any number of digits of s and concatenate them, in any order, into a positive integer. Note: Little R may choose equal digits of s, but every digit (character) of s may be used at most once. For example, if s is `1a01b`, then Little R may choose the 1st, 3rd and 4th characters simultaneously, which are $1, 0, 1$ respectively, and form the positive integer 101 or the positive integer 110; but Little R cannot form the positive integer 111, because s contains only two digits 1. Little R wants to know which of all the positive integers he can form is the largest. You need to help Little R find the maximum positive integer he can form.

Input

The first line of input contains a string s, the string Little X gave to Little R.

Output

Output one line with one positive integer, the maximum positive integer that Little R can form.

Sample 1 input

```
5
```

Sample 1 output

```
5
```

Sample 2 input

```
290es1q0
```

Sample 2 output

```
92100
```

Explanation of sample 2

The string s contains the digits $2, 9, 0, 1, 0$. It can be shown that the maximum positive integer Little R can form is 92100.

Sample 3

Large sample, not printed here; provided as the downloadable files `number/number3.in` and `number/number3.ans`. This sample satisfies the constraints of test cases $9 \sim 11$.

Sample 4

Large sample, not printed here; provided as the downloadable files `number/number4.in` and `number/number4.ans`. This sample satisfies the constraints of test case 20.

Constraints

Let $|s|$ denote the length of the string s. For all test data it is guaranteed that:

- $1 \leq |s| \leq 10^6$;
- s consists only of lowercase English letters and digits, and contains at least one digit among $1 \sim 9$.

Test case	$\|s\| \leq$	Special property
1	1	A
2	2	as above
3	as above	none
4	10	A
$5, 6$	as above	none
$7, 8$	10^2	A
$9 \sim 11$	as above	none
12	10^3	A
$13, 14$	as above	none
15	10^5	A
$16, 17$	as above	B
$18, 19$	as above	none
20	10^6	A
$21, 22$	as above	B
$23 \sim 25$	as above	none

- Special property A: s consists only of digits.
- Special property B: s contains at most 10^3 digits.

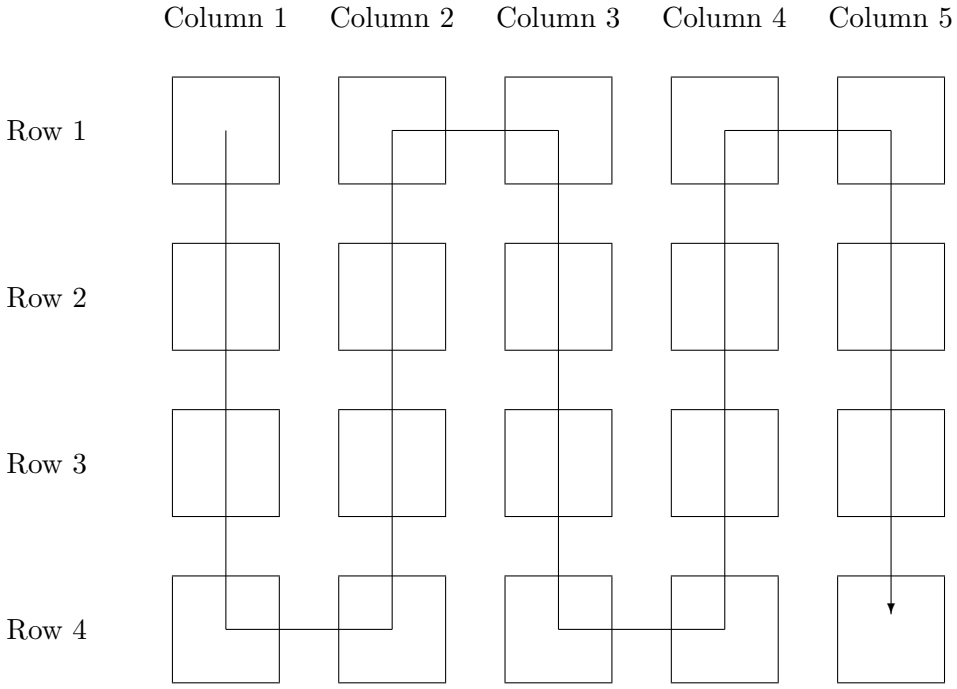
Problem 2: Seats (座位)

Luogu P14358

Time limit: 1.0 s. Memory limit: 512 MB.

The second round of CSP-J 2025 is in progress. The examination room of Little R has $n \times m$ contestants in total, and the CSP-J 2025 first-round scores of all contestants are pairwise distinct. All $n \times m$ contestants are assigned seats according to their CSP-J 2025 first-round scores, from high to low in a snake-like order, arranged in n rows and m columns. Concretely, let the scores of all contestants in Little R's examination room, from high to low, be $s_1 > s_2 > \cdots > s_{n \times m}$. Then the contestant with score s_1 sits in column 1, row 1; the contestant with score s_2 sits in column 1, row 2; ...; the contestant with score s_n sits in column 1, row n; the contestant with score s_{n+1} sits in column 2, row n; ...; the contestant with score s_{2n} sits in column 2, row 1; the contestant with score s_{2n+1} sits in column 3, row 1; and so on.

For example, if $n = 4, m = 5$, then all $4 \times 5 = 20$ contestants are assigned seats in decreasing order of their CSP-J 2025 first-round scores, following the arrows in the figure below.



Given the number of rows n and the number of columns m of the seats in Little R's examination room, together with the CSP-J 2025 first-round scores $a_1, a_2, \ldots, a_{n \times m}$ of all contestants in Little R's examination room, where a_1 is the CSP-J 2025 first-round score of Little R, you need to help Little R find out in which column and which row his seat is.

Input

The first line of input contains two positive integers n, m, the number of rows and the number of columns of the seats in Little R's examination room.

The second line of input contains $n \times m$ positive integers $a_1, a_2, \ldots, a_{n \times m}$, the CSP-J 2025 first-round scores of all contestants in Little R's examination room, where a_1 is the CSP-J 2025 first-round score of Little R.

Output

Output one line with two positive integers c, r, meaning that Little R's seat is in column c, row r.

Sample 1 input

```
2 2
99 100 97 98
```

Sample 1 output

```
1 2
```

Explanation of sample 1

In decreasing order of score, the contestant with score 100 sits in column 1, row 1; the contestant with score 99 sits in column 1, row 2; the contestant with score 98 sits in column 2, row 2; the contestant with score 97 sits in column 2, row 1. Little R's score is 99, so his seat is in column 1, row 2.

Sample 2 input

```
2 2
98 99 100 97
```

Sample 2 output

```
2 2
```

Explanation of sample 2

In decreasing order of score, the contestant with score 100 sits in column 1, row 1; the contestant with score 99 sits in column 1, row 2; the contestant with score 98 sits in column 2, row 2; the contestant with score 97 sits in column 2, row 1. Little R's score is 98, so his seat is in column 2, row 2.

Sample 3 input

```
3 3
94 95 96 97 98 99 100 93 92
```

Sample 3 output

```
3 1
```

Constraints

For all test data it is guaranteed that:

- $1 \leq n \leq 10$, $1 \leq m \leq 10$;
- for all $1 \leq i \leq n \times m$ we have $1 \leq a_i \leq 100$, and $a_1, a_2, \ldots, a_{n \times m}$ are pairwise distinct.

Test case	$n \leq$	$m \leq$	Special property
1	1	1	AB
$2, 3$	as above	10	none
$4, 5$	10	1	as above
6	2	2	A
7	as above	as above	B
$8, 9$	as above	as above	none
10	as above	10	A
11	as above	as above	B
$12 \sim 14$	as above	as above	none
$15 \sim 17$	10	2	as above
$18 \sim 20$	as above	10	as above

Special property A: for all $1 \leq i \leq n \times m$ we have $a_i = i$.
Special property B: for all $1 \leq i \leq n \times m$ we have $a_i = n \times m - i + 1$.

Problem 3: XOR Sum (异或和)

Luogu P14359

Time limit: 1.0 s. Memory limit: 512 MB.

Little R has a sequence of n non-negative integers $a_1, a_2, \ldots, a_n$. The weight of an interval $[l, r]$ $(1 \le l \le r \le n)$ is defined as the bitwise XOR of $a_l, a_{l+1}, \ldots, a_r$, that is, $a_l \oplus a_{l+1} \oplus \cdots \oplus a_r$, where $\oplus$ denotes bitwise exclusive or in binary.

Little X gives Little R a non-negative integer k. Little X wants Little R to choose as many pairwise disjoint intervals of the sequence as possible, such that the weight of every chosen interval equals k. Two intervals $[l_1, r_1], [l_2, r_2]$ intersect if and only if they both contain at least one common index, that is, there exists $1 \le i \le n$ with $l_1 \le i \le r_1$ and $l_2 \le i \le r_2$.

For example, for the sequence $[2, 1, 0, 3]$, if $k = 2$, then Little R may choose the interval $[1, 1]$ and the interval $[2, 4]$, whose weights are 2 and $1 \oplus 0 \oplus 3 = 2$ respectively; if $k = 3$, then Little R may choose the interval $[1, 2]$ and the interval $[4, 4]$, whose weights are $1 \oplus 2 = 3$ and 3 respectively.

You need to help Little R find the maximum number of intervals he can choose.

Input

The first line of input contains two non-negative integers n, k, the length of Little R's sequence and the non-negative integer that Little X gave to Little R.

The second line of input contains n non-negative integers $a_1, a_2, \ldots, a_n$, Little R's sequence.

Output

Output one line with one non-negative integer, the maximum number of intervals that Little R can choose.

Sample 1 input

```
4 2
2 1 0 3
```

Sample 1 output

```
2
```

Explanation of sample 1

Little R may choose the interval $[1, 1]$ and the interval $[2, 4]$, whose XOR sums are 2 and $1 \oplus 0 \oplus 3 = 2$ respectively. It can be shown that the maximum number of intervals Little R can choose is 2.

Sample 2 input

```
4 3
2 1 0 3
```

Sample 2 output

```
2
```

Explanation of sample 2

Little R may choose the interval $[1, 2]$ and the interval $[4, 4]$, whose XOR sums are $1 \oplus 2 = 3$ and 3 respectively. It can be shown that the maximum number of intervals Little R can choose is 2.

Sample 3 input

```
4 0
2 1 0 3
```

Sample 3 output

```
1
```

Explanation of sample 3

Little R may choose the interval $[3, 3]$, whose XOR sum is 0. It can be shown that the maximum number of intervals Little R can choose is 1. Note: Little R cannot choose the interval $[3, 3]$ and the interval $[1, 4]$ at the same time, because both intervals contain the index 3.

Sample 4

Large sample, not printed here; provided as the downloadable files `xor/xor4.in` and `xor/xor4.ans`. This sample satisfies the constraints of test cases $4, 5$.

Sample 5

Large sample, not printed here; provided as the downloadable files `xor/xor5.in` and `xor/xor5.ans`. This sample satisfies the constraints of test cases $9, 10$.

Sample 6

Large sample, not printed here; provided as the downloadable files `xor/xor6.in` and `xor/xor6.ans`. This sample satisfies the constraints of test cases $14, 15$.

Constraints

For all test data it is guaranteed that:

- $1 \leq n \leq 5 \times 10^5$, $0 \leq k < 2^{20}$;
- for all $1 \leq i \leq n$ we have $0 \leq a_i < 2^{20}$.

Test case	$n \leq$	k	Special property
1	2	$= 0$	A
2	10	≤ 1	B
3	10^2	$= 0$	A
$4, 5$	as above	≤ 1	B
$6 \sim 8$	as above	≤ 255	C
$9, 10$	10^3	as above	as above
$11, 12$	as above	$< 2^{20}$	none
13	2×10^5	≤ 1	B
$14, 15$	as above	≤ 255	C
16	as above	$< 2^{20}$	none
17	5×10^5	≤ 255	C
$18 \sim 20$	as above	$< 2^{20}$	none

Special property A: for all $1 \leq i \leq n$ we have $a_i = 1$.
Special property B: for all $1 \leq i \leq n$ we have $0 \leq a_i \leq 1$.
Special property C: for all $1 \leq i \leq n$ we have $0 \leq a_i \leq 255$.

Problem 4: Polygon (多边形) Luogu P14360

Time limit: 1.0 s. Memory limit: 512 MB.

Little R likes playing with small sticks. Little R has n sticks, and the i-th ($1 \le i \le n$) stick has length a_i.

Little X wants Little R to choose some of these n sticks and join them end to end, in any order, into a polygon. Little R does not know the condition under which sticks can form a polygon, so Little X tells him the condition directly: m sticks of lengths $l_1, l_2, \ldots, l_m$ can form a polygon if and only if $m \ge 3$ and the sum of the lengths of all the sticks is strictly greater than twice the maximum length of the sticks, that is, $\sum_{i=1}^{m} l_i > 2 \times \max_{i=1}^{m} l_i$.

Now that Little R knows the condition for sticks to form a polygon, Little X poses a harder question: how many ways are there to choose sticks such that the chosen sticks can form a polygon? You need to help Little R count the number of ways to choose sticks such that the chosen sticks can form a polygon. Two ways are different if and only if the sets of indices of the chosen sticks are different, that is, there exists $1 \le i \le n$ such that one of the ways chooses the i-th stick but the other does not. Since the answer may be large, you only need to output the answer modulo $998,244,353$.

Input

The first line of input contains a positive integer n, the number of Little R's sticks.

The second line of input contains n positive integers $a_1, a_2, \ldots, a_n$, the lengths of Little R's sticks.

Output

Output one line with one non-negative integer, the number of ways to choose sticks such that the chosen sticks can form a polygon, modulo $998,244,353$.

Sample 1 input

```
5
1 2 3 4 5
```

Sample 1 output

```
9
```

Explanation of sample 1

There are the following 9 ways to choose sticks such that the chosen sticks can form a polygon:

1. choose sticks $2, 3, 4$; the total length is $2 + 3 + 4 = 9$ and the maximum length is 4;
2. choose sticks $2, 4, 5$; the total length is $2 + 4 + 5 = 11$ and the maximum length is 5;
3. choose sticks $3, 4, 5$; the total length is $3 + 4 + 5 = 12$ and the maximum length is 5;
4. choose sticks $1, 2, 3, 4$; the total length is $1 + 2 + 3 + 4 = 10$ and the maximum length is 4;
5. choose sticks $1, 2, 3, 5$; the total length is $1 + 2 + 3 + 5 = 11$ and the maximum length is 5;

6. choose sticks $1, 2, 4, 5$; the total length is $1 + 2 + 4 + 5 = 12$ and the maximum length is 5;
7. choose sticks $1, 3, 4, 5$; the total length is $1 + 3 + 4 + 5 = 13$ and the maximum length is 5;
8. choose sticks $2, 3, 4, 5$; the total length is $2 + 3 + 4 + 5 = 14$ and the maximum length is 5;
9. choose sticks $1, 2, 3, 4, 5$; the total length is $1 + 2 + 3 + 4 + 5 = 15$ and the maximum length is 5.

Sample 2 input

```
5
2 2 3 8 10
```

Sample 2 output

```
6
```

Explanation of sample 2

There are the following 6 ways to choose sticks such that the chosen sticks can form a polygon:

1. choose sticks $1, 2, 3$; the total length is $2 + 2 + 3 = 7$ and the maximum length is 3;
2. choose sticks $3, 4, 5$; the total length is $3 + 8 + 10 = 21$ and the maximum length is 10;
3. choose sticks $1, 2, 4, 5$; the total length is $2 + 2 + 8 + 10 = 22$ and the maximum length is 10;
4. choose sticks $1, 3, 4, 5$; the total length is $2 + 3 + 8 + 10 = 23$ and the maximum length is 10;
5. choose sticks $2, 3, 4, 5$; the total length is $2 + 3 + 8 + 10 = 23$ and the maximum length is 10;
6. choose sticks $1, 2, 3, 4, 5$; the total length is $2 + 2 + 3 + 8 + 10 = 25$ and the maximum length is 10.

Sample 3

Large sample, not printed here; provided as the downloadable files `polygon/polygon3.in` and `polygon/polygon3.ans`.

This sample satisfies the constraints of test cases $7 \sim 10$.

Sample 4

Large sample, not printed here; provided as the downloadable files `polygon/polygon4.in` and `polygon/polygon4.ans`.

This sample satisfies the constraints of test cases $11 \sim 14$.

Constraints

For all test data it is guaranteed that:

- $3 \leq n \leq 5\,000$;
- for all $1 \leq i \leq n$ we have $1 \leq a_i \leq 5\,000$.

Test case	$n \leq$	$\max_{i=1}^{n} a_i \leq$
$1 \sim 3$	3	10
$4 \sim 6$	10	10^2
$7 \sim 10$	20	as above
$11 \sim 14$	500	as above
$15 \sim 17$	as above	1
$18 \sim 20$	$5\,000$	as above
$21 \sim 25$	as above	$5\,000$

Source

The statements were translated from the official problems as published on Luogu (CSP-J 2025, second round, held on 2025-10-25):

- Problem 1: `https://www.luogu.com.cn/problem/P14357`
- Problem 2: `https://www.luogu.com.cn/problem/P14358`
- Problem 3: `https://www.luogu.com.cn/problem/P14359`
- Problem 4: `https://www.luogu.com.cn/problem/P14360`